Herald College, Kathmandu

Artificial Intelligence and Machine Learning

6CS012 — Final Portfolio Assessment

Part I — Question and Answer (QnA)

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April, 2026

Long Question — Answer 1 of 2 [5 Marks]

Question 1: You are a Data Scientist at eSewa, a leading digital payment platform, working on large-scale user transaction data. Identify two high-impact business problems where unsupervised learning can provide value. For each problem, define it in context, propose a suitable approach, discuss system integration, and briefly mention one limitation.

Problem 1: Account Takeover Detection via Session Behaviour Analysis

Business Context. eSewa, as a digital wallet, is a primary target for attackers. Malware, session hijacking, or phishing can grant full access to a victim's account. The real problem is not the login itself — it is what the attacker does afterward.

Proposed Approach: LSTM Autoencoder. User sessions are sequential by nature. There is order, rhythm, and purpose in legitimate user actions. A standard autoencoder compresses a session to a vector, losing all temporal information. LSTM cells fix this: the input gate controls what new information enters the cell state, the forget gate decides what to discard, and the output gate determines what carries forward. The model learns not just individual actions but the transitions between them. A sequence like 'balance check → small transfer → logout' reconstructs cleanly. A sequence like 'login → immediate maximum transfer → add new beneficiary → logout' carries high reconstruction error, because those transitions never appeared in training on legitimate sessions. The model is trained exclusively on normal sessions. At inference, an attacker navigating unfamiliar paths or skipping steps a real user would never skip accumulates reconstruction error that silently triggers a mid-session OTP re-verification — without alerting the attacker. Features used: navigation sequences, time per screen, input cadence, and transaction speed.

System Integration. Sessions stream through RabbitMQ. Scoring happens continuously — not just at login. When the anomaly score crosses a threshold, silent OTP re-verification is triggered before any fund movement proceeds.

Limitation. Users on new devices or in unfamiliar contexts generate legitimate false positives, which means per-user threshold calibration is necessary rather than a single global cutoff.

Problem 2: API Abuse and Enumeration Attack Detection

Business Context. eSewa's APIs for account verification, merchant lookup, and balance checking are probed by attackers to map valid accounts or scrape merchant data. These requests are intentionally slow and stay under rate limits, making them invisible to threshold-based detection.

Proposed Approach: LSTM Autoencoder on API Call Sequences. Legitimate API usage has structure: calls follow logical progressions tied to real user intent — authenticate → fetch balance → initiate payment → confirm. The LSTM Autoencoder learns this temporal grammar. The encoder compresses a session's API call sequence into a latent vector representing behavioural intent. The decoder reconstructs the original sequence. Legitimate sessions reconstruct with low error. Enumeration attacks are structurally empty — the same endpoints are called with incrementally different parameters, with no logical flow. The forget gate means the LSTM never accumulates meaningful hidden state from these repetitive patterns. Reconstruction error stays consistently high, which is itself a signal — legitimate session error typically decreases as the sequence follows its predicted progression. Features: endpoint identifiers, inter-call intervals, parameter variability between consecutive calls, and response code patterns.

System Integration. Runs on a 5-minute session window as a near-real-time batch job. Flagged clients are throttled silently — responses slow down slightly rather than hard-blocking, so attackers remain unaware of detection.

Limitation. Third-party integrations and partner merchant reconciliation jobs have structurally repetitive API patterns similar to enumeration. Without upstream whitelisting of known automated clients, these will generate a constant stream of false positives.

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Short Questions — Pick 1 from Each Category [5 Marks]

Category 1: Overfitting [2.5 Marks]

Question 2: Overfitting is a common challenge in deep learning, especially when models have high capacity. Describe at least two techniques used to reduce overfitting, explaining the intuition behind each, how it affects training, and a real-world application example.

Technique 1: Dropout

When a neural network has more capacity than the training data it is trained on, it memorises the training data rather than learning general patterns — that is overfitting. Dropout prevents this by randomly disabling a fraction of neurons during each training step. The network can never rely on any specific neuron, which forces it to learn distributed, robust representations. In the testing phase all neurons are re-enabled, but their outputs are scaled to compensate for the dropout applied during training. For image classification in KYC validation, dropout prevents the model from latching onto specific pixel regions, instead compelling it to recognise documents from multiple overlapping cues rather than a few isolated details.

Technique 2: Data Augmentation

Sometimes a model overfits not because it is poorly designed but because the training data is too small or not diverse enough. Data augmentation expands the effective training set by generating transformed versions of existing examples — flipping, rotating, cropping, adjusting brightness, or adding noise — without collecting new data. The model now encounters slightly different versions of each example, making memorisation much harder and generalisation much easier. For training a cheque or document scanner with limited samples, augmenting with rotations, brightness shifts, and blur simulates realistic scanning conditions and dramatically improves real-world performance.

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Category 2: Neural Network Architecture [2.5 Marks]

Question 2: FCN vs. Autoencoder — Compare a standard feedforward neural network (FCN) with an autoencoder in terms of architecture, objective, and input-output relationship. Provide an example and describe one application of autoencoders.

Architecture

A fully connected network flows from input through hidden layers to an output layer in one direction, with every neuron connected to every neuron in the next layer. An autoencoder shares this structure but inserts a bottleneck: an encoder compresses the input into a smaller latent representation, and a decoder reconstructs the original input from that compressed form.

Objective and Functionality

An FCN uses supervised learning — it maps inputs to predefined labels. An autoencoder is unsupervised — it has no external labels. Its only objective is to reconstruct its own input as accurately as possible, which forces it to learn the most essential structure in the data.

Input-Output Relationship

In an FCN, input and output are fundamentally different things: an image goes in, a class label comes out. In an autoencoder, input and output are the same thing: an image goes in and a reconstructed version of the same image (or a denoised version) comes out.

Example and Application

Consider input vector [0.9, 0.1, 0.8, 0.95]. An FCN outputs a class prediction ('cat' or 'dog'). An autoencoder compresses and decompresses it, producing denoised output like [0.98, 0.02, 0.97, 0.99]. In a real KYC pipeline, an autoencoder trained only on legitimate citizenship scans learns the structural and pixel-level patterns of genuine documents. When a forged document passes through, the decoder cannot reconstruct it accurately — the latent vector does not capture the unfamiliar patterns of the forgery. This elevated reconstruction error flags the document as anomalous without ever needing labelled examples of fake documents.

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